

ASSIGNMENT 4.1

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Scenario	How the Multimeter Helps
1. laptop not charging	It can measure the voltage from the laptop charger and charging circuit to check whether the correct voltage is reaching the laptop.
2. desktop PC not turning on	It can check the power supply voltages to determine whether the PSU is providing proper power.
3. smartphone not charging	It can be used to check voltage and continuity in the charging port or related circuit to identify a faulty connection or component.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

- A POST card is used to diagnose hardware problems during the computer's startup process. It displays POST codes generated by the motherboard while checking hardware components. In the demonstration video, the POST card displayed a specific error code, which indicated a hardware-related problem. The code helped the technician identify the faulty components and troubleshoot the motherboard.

3.Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or Crystal Disk Info), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

CPUID HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
DHEW3GS				
Intel Core i5				
Voltages				
CPU VCORE	0.672 V	0.648 V	1.152 V	
+12V	12.096 V	12.096 V	12.192 V	
+5V	5.040 V	5.000 V	5.080 V	
+3.3V	3.312 V	3.296 V	3.328 V	
Temperatures				
Package	42 °C (107 °F)	40 °C (104 °F)	78 °C (172 °F)	
Cores (Max)	40 °C (104 °F)	38 °C (100 °F)	75 °C (167 °F)	
Powers				
Package	12.34 W	8.21 W	45.76 W	
IA Cores	8.12 W	4.87 W	38.22 W	
GT	1.87 W	1.32 W	8.45 W	
Utilization				
Processor	5 %	1 %	100 %	
CPU #0	6 %	0 %	100 %	
CPU #1	3 %	0 %	100 %	
CPU #2	4 %	0 %	100 %	
CPU #3	7 %	0 %	100 %	
Clocks				
Core #0	2394 MHz	798 MHz	4090 MHz	
Core #1	2394 MHz	798 MHz	4090 MHz	
Core #2	2494 MHz	798 MHz	4090 MHz	
Core #3	2494 MHz	798 MHz	4090 MHz	
NVIDIA GeForce GTX 1650				
Temperatures				
GPU	45 °C (113 °F)	40 °C (104 °F)	72 °C (161 °F)	
Utilization				
GPU	3 %	0 %	99 %	
Memory	12 %	8 %	62 %	
Clocks				
Graphics	300 MHz	300 MHz	1800 MHz	
Memory	405 MHz	405 MHz	6001 MHz	
STORAGE (SSD)				
Temperatures				
Assembly	32 °C (89 °F)	31 °C (87 °F)	45 °C (113 °F)	
Ready				
				DHEW3GS

4.Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example problem detected
Multimeter	Checks voltage, resistance, and continuity in electrical circuits.	Faulty power supply or damaged charging circuit
POST card	Checks the motherboard’s startup process and displays POST error codes.	RAM, CPU, or motherboard hardware failure
Software Diagnostics	Checks hardware information, temperature, storage health, and system performance.	CPU overheating or failing HDD/SSD